# IBVAP - AI Engine Guide

Person A ke liye poora kaam: setup, kya banana hai, backend se judna, Claude Code prompts, plan

## 1. Tumhara kaam ek line mein

**Camera ka video lo, usme AI chalao, aur jo mila uska alert backend ko bhejo.** Dashboard (frontend) already ban chuka hai. Backend (FastAPI + database) hum dono milke banayenge. Tumhara folder hai `ai_engine/`.

| Part | Kaun | Status |
|---|---|---|
| Frontend dashboard (React) | Person B | Ban chuka hai (mock data pe chal raha) |
| Backend (FastAPI + SQLite) | Dono milke | Banana hai |
| AI engine (detection, tracking, fence, ANPR, face) | Tum (Person A) | Banana hai |

**Sabse important rule:** tumhara AI engine **bina backend ke bhi chalna chahiye** (standalone mode: video window mein boxes dikhaye aur alerts ek local file mein likhe). Isse tum backend ka wait kiye bina Day 1 se kaam kar sakte ho.

## 2. Kya-kya banana hai (priority order)

| # | Module | Kya karega | Priority |
|---|---|---|---|
| 1 | Video input | Demo video file, webcam, RTSP ya phone IP Webcam padhna. 2-4 cameras ek saath | Must |
| 2 | Detection | YOLO se person, car, truck, bus, motorcycle. Jaanwar (dog, cow, horse, sheep) alag pehchanna, unpe alert nahi | Must |
| 3 | Tracking | ByteTrack (supervision) se har object ko stable track ID | Must |
| 4 | Virtual fence | Polygon zone + tripwire line. Object kuch frames zone mein rahe tabhi alert (false alarm kam) | Must |
| 5 | Alerts | Har alert: time, camera, type, severity, confidence, track ID, bbox, snapshot, details | Must |
| 6 | ANPR | Gaadi pe plate dhoondhna, EasyOCR se padhna, Indian format check, blacklist match | Must |
| 7 | Backend se judna | Zones/blacklist backend se lena, alerts backend ko bhejna, boxes wala video stream dena | Must |
| 8 | Loitering | Koi zone mein X second se zyada ruke | Should |
| 9 | Night mode | Andhere frame pe CLAHE brightness, raat mein alag threshold, NIGHT_MOVEMENT alert | Should |
| 10 | Face | Face detection pakka; watchlist match (recognition) bonus | Bonus |

## 3. Setup (Day 1, pehla 1 ghanta)

**Install karo:** Python 3.10 ya 3.11 (3.12 pe kuch libraries dikkat deti hain), Git, VS Code, Claude Code.

```
git clone <team repo url>
cd ibvap
mkdir ai_engine
cd ai_engine
python -m venv .venv
# Windows: .venv\Scripts\activate
# Mac/Linux: source .venv/bin/activate
pip install ultralytics supervision opencv-python easyocr numpy requests pyyaml fastapi uvicorn
```

**Face ke liye (baad mein, Day 5):** `pip install insightface onnxruntime`. Windows pe install fail ho to face ko sirf detection tak rakho (section 9 dekho).

**Demo videos:** `ai_engine/demo_videos/` mein rakho (git mein mat daalna, size bada hota hai). Chahiye:

- Log chalte hue, ek line/area cross karte hue (fence demo) - khud phone se record karo, best
- Gaadiyan jinki number plate saaf dikhe (parking ya gate, din ka time)
- Raat ka video (kam roshni mein koi chal raha ho)
- Optional: koi ek jagah khada rahe 30+ second (loitering)
- Free source: Pexels.com, Pixabay.com. YouTube videos mat use karna (copyright)

## 4. Folder structure

```
ai_engine/
  config.yaml cameras list, backend URL, thresholds, mode (standalone/backend)
  main.py sab start karta hai (har camera ka ek thread)
  camera.py video/RTSP padhna, reconnect, frame skip
  detector.py YOLO wrapper (person/vehicle/animal classes)
  tracker.py ByteTrack (supervision)
  rules/
    fence.py polygon + tripwire, min frames, direction
    loitering.py dwell time per track
    night.py dark frame check + CLAHE
  anpr/
    plate_detector.py plate box (Roboflow pretrained model / YOLO plate weights)
    ocr.py EasyOCR + cleanup + Indian format regex
  face/
    face.py detection + watchlist embeddings match
  alerts.py alert banana, cooldown/dedupe, snapshot save
  backend_client.py GET zones/blacklist/watchlist, POST alerts (retry + offline queue)
  stream_server.py MJPEG stream per camera (boxes drawn) on port 8001
  models/ downloaded weights (git ignore)
  demo_videos/ (git ignore)
  output/ standalone mode: alerts.jsonl + snapshots/
  requirements.txt
  README.md
```

## 5. Backend se kaise judna hai (contract)

Frontend ke paas `frontend/docs/API_CONTRACT.md` hai. **Pehle ise poora padho.** Alert ka format, alert types aur severity ke naam wahi use karne hain, warna dashboard pe kuch nahi dikhega.

| Direction | Kya | Kaise |
|---|---|---|
| Backend -> AI | Cameras, zones, blacklist, watchlist | AI engine har 10-30 sec GET karke refresh kare (zones dashboard ke Zones editor se badalte hain) |
| AI -> Backend | Alerts + plate reads | POST JSON (snapshot base64 ya multipart). Backend fail ho to local queue, baad mein retry |
| AI -> Dashboard | Boxes wala live video | MJPEG stream: http://localhost:8001/stream/{camera_id} (dashboard isko img src mein dikhata hai) |
| AI -> Backend | Live detections (optional) | Har camera ke boxes ka chhota JSON, overlay ke liye |
| AI -> Backend | Health | FPS, latency, CPU/GPU har 10 sec (System Health page ke liye) |

**Contract ke important rules:**

- Alert types: FENCE_INTRUSION, HUMAN_DETECTED, VEHICLE_DETECTED, ANPR_MATCH, FACE_MATCH, LOITERING, NIGHT_MOVEMENT, SUSPICIOUS_ACTIVITY, CAMERA_OFFLINE
- Severity: CRITICAL, HIGH, MEDIUM, LOW
- bbox aur zone points **normalized 0-1** mein (pixel nahi), taaki har resolution pe sahi bane
- Time ISO format mein, IST
- Agar AI engine ke liye koi endpoint contract mein nahi hai (jaise POST alert), to Person B se baat karke contract mein add karo, phir banao

## 6. Alert banane ke rules (false alarm kam karne ke liye)

- **Min frames:** zone mein object kam se kam 5-8 consecutive processed frames tak dikhe tabhi alert
- **Cooldown:** ek hi track ID + zone pe 60 sec tak dobara alert nahi
- **Animals ignore:** dog, cow, horse, sheep, bird pe fence alert nahi
- **Confidence:** din 0.5, raat 0.35-0.4 (raat mein model kam confident hota hai)
- **Tripwire direction:** sirf ek taraf se cross hone pe (jaise border side se andar)
- **Har alert ke details mein:** kitne frames confirm hua, dwell time, zone naam, class - dashboard ye dikhata hai
- **ANPR:** same plate ek camera pe 2 min mein ek hi baar; OCR confidence kam ho to bhi bhejo par 'verify' flag ke saath

## 7. Claude Code prompts (ek-ek karke do)

Claude Code repo ke main folder `ibvap/` mein chalao (taaki wo frontend ka contract padh sake). Har prompt ke baad khud chala ke check karo, commit karwao, phir `/clear` aur agla prompt.

### AI-1: Setup + detection + tracking (Day 1)

```
We are building the AI engine for IBVAP (Intelligent Border Video Analytics Platform, Smart India Hackathon PS 26187). It turns existing CCTV cameras at border posts into a smart surveillance system. The React dashboard is already built in frontend/. Read frontend/CLAUDE.md and frontend/docs/API_CONTRACT.md first so you understand the data formats.

Create the AI engine in ai_engine/ (Python 3.10/3.11). Do this first task only:
1. Create the folder structure: config.yaml, main.py, camera.py, detector.py, tracker.py, rules/, anpr/, face/, alerts.py, backend_client.py, stream_server.py, models/, output/, requirements.txt, README.md. Add models/, demo_videos/, output/ and .venv/ to .gitignore.
2. config.yaml: mode (standalone | backend), backend_url, list of cameras (id, name, source = video file path / webcam index / RTSP or HTTP URL, lat, lng), detection settings (model yolov8n or yolo11n, imgsz 640, process every Nth frame, confidence day/night).
3. camera.py: one thread per camera, reads frames with OpenCV, loops demo video files, auto-reconnects RTSP with backoff, keeps only the latest frame (no lag buildup), reports FPS.
4. detector.py: Ultralytics YOLO, classes person, car, truck, bus, motorcycle, bicycle, plus animals (dog, cow, horse, sheep, bird) tagged as animal. Use GPU if available, else CPU.
5. tracker.py: ByteTrack via the supervision library, stable track IDs per camera.
6. main.py: runs all cameras; in standalone mode shows an OpenCV window per camera with boxes, class, track ID, confidence and FPS. Press q to quit.
7. README.md: how to set up the venv, install, put demo videos, and run.
Run it on a sample video if available and fix all errors. Use Indian border style camera names in config (e.g. "BOP Sundar - Gate Cam", "Border Road KM 42").
```

## AI-2: Virtual fence + loitering + alerts (Day 2)

```
Read ai_engine/README.md and frontend/docs/API_CONTRACT.md. Now add rules and alerts:
1. rules/fence.py: polygon zones and tripwire lines per camera, points normalized 0-1 (same shape as the Zone type in the contract). Use supervision PolygonZone / LineZone. Tripwire supports direction (only alert when crossing in the configured direction). Zone has: active, schedule (always / night-only), classes that trigger it, severity, min dwell seconds.
2. Alert only when a track stays in the zone for at least N processed frames (config, default 6). Animals never trigger fence alerts.
3. rules/loitering.py: LOITERING alert when a person stays inside a zone longer than loitering_seconds (config, default 30).
4. alerts.py: builds alert objects EXACTLY in the Alert format of API_CONTRACT.md (type names, severity names, normalized bbox, ISO time in IST, trackId, confidence, details with zone name, dwell time, frames confirmed). Saves a snapshot JPG with the box drawn. Cooldown: same track + zone + type not repeated within 60 seconds.
5. Standalone mode: append alerts to output/alerts.jsonl and snapshots to output/snapshots/, print a one-line summary, and draw zones (semi-transparent) on the OpenCV window; flash the border red on alert.
6. Until the backend exists, load zones from a local zones.json (2-3 sample zones per camera). Add a tiny helper script to draw a zone by clicking on the first frame of a video and save it to zones.json.
Test on a video where people cross a line and fix all errors.
```

## AI-3: Backend connection + MJPEG stream (Day 3-4, backend ready hone ke baad)

```
The FastAPI backend now exists in backend/. Read backend code and frontend/docs/API_CONTRACT.md.
1. backend_client.py: in backend mode, fetch cameras, zones, blacklist and watchlist from the backend on start and refresh every 15 seconds (zones edited in the dashboard must take effect without restart). POST every alert and plate read to the backend in the contract format. If the backend is down, queue to output/queue.jsonl and retry with backoff; never crash the engine.
2. If an endpoint the engine needs is missing from the contract (e.g. ingesting alerts from the engine, health), add it to API_CONTRACT.md AND implement it in backend/ to match, keeping the existing frontend endpoints unchanged.
3. stream_server.py: a small FastAPI/uvicorn server on port 8001 serving MJPEG at /stream/{camera_id} with boxes, track IDs and zones drawn, and /snapshot/{camera_id} for a single JPG. Make sure CORS allows the dashboard.
4. Send camera health (FPS, latency, status online/offline) to the backend every 10 seconds; emit a CAMERA_OFFLINE alert when a source stops for 30 seconds.
5. main.py: one command starts the engine + stream server.
Test end to end: run backend, run engine, run the dashboard with VITE_USE_MOCK=false, and confirm a real fence alert appears live on the dashboard. Fix any mismatch.
```

## AI-4: ANPR (Day 4)

```
Add ANPR to the AI engine (ai_engine/anpr/):
1. plate_detector.py: detect licence plates inside vehicle boxes using a pretrained YOLO licence-plate model (download weights from Roboflow Universe or Hugging Face; put the path in config.yaml and explain in README how to download it). Only run it every few frames per tracked vehicle, not on every frame.
2. ocr.py: crop the plate, upscale, grayscale, CLAHE, run EasyOCR (English). Clean the text, fix common OCR confusions (O/0, I/1, B/8, S/5) by position, and validate with an Indian plate regex (e.g. RJ14AB1234, also BH series). Keep the best read per track across frames (vote), with a confidence score.
3. Send a plate read for every vehicle track (format from the contract: plate, vehicle class, camera, time, confidence, crop snapshot). If confidence is low mark it for verification.
4. If the plate is in the blacklist (from backend or local blacklist.json in standalone), create an ANPR_MATCH alert with severity from the blacklist entry.
5. Same plate on the same camera is reported at most once per 2 minutes.
Test on a parking/gate video with clear plates and show me sample reads.
```

### AI-5: Night mode + face (Day 5)

```
Add to the AI engine:
1. rules/night.py: detect dark frames (mean brightness below threshold) per camera; when dark,
apply CLAHE enhancement before detection, use the night confidence threshold, mark the camera
isNightMode=true in health, and emit NIGHT_MOVEMENT alerts for persons/vehicles in night-only
zones.
2. face/face.py: face detection on person crops (InsightFace if it installs; if not, fall back to
OpenCV DNN/YuNet face detector and skip recognition). Recognition: build embeddings from watchlist
photos (from backend or a local watchlist/ folder, one sub-folder per person), cosine similarity,
FACE_MATCH alert above threshold (config, default 0.45) with similarity score, captured face crop
and matched person id. Only check each track a few times, not every frame.
3. Add a README note: face match is decision support only; final identification by authorised
personnel.
Make face optional via config so the engine still runs if face libraries are missing. Test and fix
errors.
```

### AI-6: Performance + demo ready (Day 6)

```
Make the AI engine demo-ready:
1. Performance: confirm it runs 3-4 cameras on a laptop CPU. Use frame skipping, imgsz 640, nano
model, run ANPR/face only on tracked objects every few frames. Print per-camera FPS and latency.
Add a "low power" profile in config.
2. Stability: run for 30 minutes on looping demo videos without memory growth or crashes; handle a
camera source disappearing and coming back.
3. Add run_demo script (run_demo.bat for Windows and run_demo.sh) that starts backend, AI engine
and frontend together.
4. Update README.md: setup, config, how to add a camera, how to download the plate model, how to
run standalone vs backend mode, troubleshooting.
5. Remove debug prints, keep clean logs. Commit.
Give me a short report of FPS per camera and anything still unreliable.
```

## 8. Din-wise plan (Person A)

| Din | Kaam | Shaam tak check |
|---|---|---|
| Day 1 | Setup, demo videos jama karna, prompt AI-1 | Video pe boxes + track ID dikh rahe |
| Day 2 | Prompt AI-2 (fence, loitering, alerts) | Line cross karne pe alerts.jsonl mein alert + snapshot |
| Day 3 | Backend dono milke (FastAPI + SQLite, contract ke hisaab se) | Backend API alerts save aur return kar rahi |
| Day 4 | Prompt AI-3 (backend se judna, MJPEG), phir AI-4 (ANPR) | Asli fence alert dashboard pe live dikh raha (sabse important) |
| Day 5 | Prompt AI-5 (night + face) | Raat ka video aur face match chal raha |
| Day 6 | Prompt AI-6, Person B ke saath poora test | Ek command se sab chalu, 30 min bina crash |
| Day 7 | Demo video record, PPT mein AI part ki slides | Submit |

**Time kam pade to:** AI-1, AI-2, AI-3 aur AI-4 zaroor. AI-5 mein sirf night mode. Face ko future scope bol do.

## 9. Common errors aur fix

| Problem | Fix |
|---|---|
| Video bahut slow (2-3 FPS) | yolov8n / yolo11n, imgsz 640 ya 480, har 3rd frame process, ANPR/face kam frames pe. Ya Google Colab pe run karke output video record karo |
| torch / ultralytics install error | Python 3.10/3.11 use karo, naya venv banao, pip upgrade: python -m pip install --upgrade pip |
| GPU use nahi ho raha | CPU pe bhi chalega. NVIDIA GPU hai to pytorch.org se CUDA wala torch install karo |

| Problem | Fix |
|---|---|
| EasyOCR pehli baar atka hua | Pehli baar model download hota hai (internet chahiye), 1-2 minute ruko |
| InsightFace / dlib install fail (Windows) | Face recognition skip karo, sirf OpenCV YuNet face detection rakho (prompt AI-5 mein fallback hai) |
| RTSP stream lag ya freeze | Sirf latest frame rakho (buffer clear), reconnect logic, ya recorded video ko MediaMTX se RTSP banao |
| Phone IP Webcam connect nahi ho raha | Phone aur laptop same WiFi pe, URL http://PHONE_IP:8080/video, firewall check |
| Dashboard pe alert nahi aa raha | Alert JSON contract se match karo (type/severity ke naam, normalized bbox), backend logs dekho, CORS check |
| Number plate galat padh raha | Clear daylight video, plate crop bada karo, multiple frames ka vote, regex se clean |
| cv2.imshow error (no display) | Standalone window ki jagah sirf MJPEG stream / output files use karo |

## 10. Kaam 'done' kab maana jayega

- Ek command se AI engine chalu, 3-4 demo videos ek saath
- Person/vehicle boxes + track ID, jaanwar pe fence alert nahi
- Fence/tripwire cross pe alert, snapshot ke saath, dashboard pe live (2-3 sec ke andar)
- Dashboard ke Zones editor mein zone badlo to AI engine bina restart naya zone use kare
- Gaadi ki plate padhi jaye, blacklist plate pe ANPR_MATCH alert
- Raat ke video pe detection kaam kare
- Backend band ho to engine crash na ho, alerts queue ho
- README se koi bhi naya banda setup kar sake

Koi bhi problem 2 ghante se zyada atke to chhod ke aage badho aur Person B ko batao. Demo ke liye 4 jude hue features, 8 adhoore features se behtar hain.